Network Forensic Report

PCAP Network Packet Capture Analysis

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This report provides the details of the forensic analysis performed on the network capture file "XYZ.pcap" in XYZ School.

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1. Executive Summary

This report demonstrates a comprehensive network forensic investigation into a harassing email incident targeting a member in XYZ academic community. The investigation was conducted using packet capture analysis on the network file XYZ.pcap. The primary objective was to recognize the identity of the person who sent the harassing messages, understand the transmission method, recover the message content where possible and to accurately associate the activity with a specific individual based on reliable forensic evidence.

Through structured analysis of captured traffic in Wireshark, it was possible to identify one of the internal devices that was responsible for a high level of network traffic. Subsequent analysis of the application layer traffic disclosed evidence about the authenticated users and an unencrypted HTTP form submission to http://www.willselfdestruct.com. Examination of this plaintext transmission enabled the extraction of the subject line and body content of the email, which contained offensive and derogatory comments about both the recipient and the class. Through the correlation of network-level evidence and institutional records, it was possible to associate the activity with the device assigned to Johnny Coach.

This report represents the full scope of the investigation, outlining the approach adopted, tools utilized, key findings, and conclusions drawn. The analysis will deliver a clear and justifiable means of attribution, complying with accepted digital forensic standards and best practices.

1. **Network Activity Analysis**
   There were multiple devices on the network that were active on the 192.168.15.xx internal network range, but the device assigned the IP address 192.168.15.4 generated a significantly higher volume of traffic than other devices on the network, which was indicative of user-driven activity.

2. **Irrelevant Devices**
   The other devices on the network were only sending out background traffic and were not involved in any communication that was relevant to the incident.

3. **Gateway Device**
   The internal gateway device, identified as 192.168.15.1, functioned as the network router. It facilitated the routing of outbound traffic from the suspect device to external devices and servers.

4. **External Communications**
   According to the information revealed by the analysis, the suspect device has established communications with multiple external servers, including willselfdestruct.com, which complies with standard internet communication architecture.

5. **System Identification**
   User agent header analysis indicated that the suspect system was a desktop computer operating a standard web browser, which was consistent with human interaction.

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6. **Authenticated Session Activity**
   An authenticated Google session on the suspect device for the user jcoachj@gmail.com, was identified on the suspect device during the timeframe of the incident. This session was associated with the observed network activity.

7. **Harassing Email Activity**
   The suspect device sent an HTTP POST request to http://www.willselfdestruct.com and the TCP stream reconstruction indicated that the email was transmitted in plaintext to lilytuckrige@yahoo.com and contained offensive and harassing messages.

8. **Attribution**
   The correlation of the suspect IP address, the authenticated session information, the message content, and the institutional information attributed the activity to Johnny Coach, a student enrolled in CHEM109.This analysis attributes the activity to the device rather than conclusively identifying the individual operator.

9. **Accomplice Assessment**
   Within the defined scope of investigation,There was no evidence of accomplices found within the scope of the investigation.

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2. Introduction

This investigation was conducted following a formal complaint from Lily Tuckrige, a Chemistry lecturer at XYZ school, who reported the Administration about the being the recipient of numerous harassing emails sent through her personal email address. An examination of the headers of the emails first showed that these emails originated from an IP address associated with a student living in a dormitory. The IT Department at XYZ School took the opportunity to use a network packet sniffer to capture all network traffic from the dormitory Ethernet port because it was a shared network by many users, and an unsecured wireless router was being used in the dormitory. The purpose of this investigation was to conduct a forensic analysis of the captured network traffic to determine who was responsible for sending the harassing email, to reconstruct the timeline of events that led up to the sending of the harassing email, and to determine if a student currently enrolled in Chemistry 109 was responsible for sending the harassing email. The conclusion reached in this report are the result of a systematic packet analysis, TCP stream reconstruction, and correlation of network artifacts to establish attribution supported by verifiable digital evidence.

2.1 Network Capture File details

The extracted PCAP network capture file XYZ.pcap has the forensic parameters as given below. The evidence for these details is provided in Figure 1 extracted from Wireshark 4.6.3 (v4.6.3-0-g648f69f3e168):

Capture length:	54,670,237 bytes
Format:	Wireshark/tcpdump/... – pcap
Packet size limit:	4096 bytes
First packet:	22-JUL-2008 07:21:07 HRS
Last packet:	22-JUL-2008 11:43:47 HRS
Elapsed time:	04 hours, 22 minutes, 39 seconds (15759.951 seconds)
Total packets:	94,410
Average packets/sec	6.0 packets/sec
Average packet size	579 bytes
Average bytes/sec	3,468 bytes/sec
Average bits/sec	27 kbit/sec

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Computed HASHes – XYZ.pcap

MD5: 9981827f11968773ff815e39f5458ec8

SHA1: 65656392412add15f93f8585197a8998aaeb50a1

SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

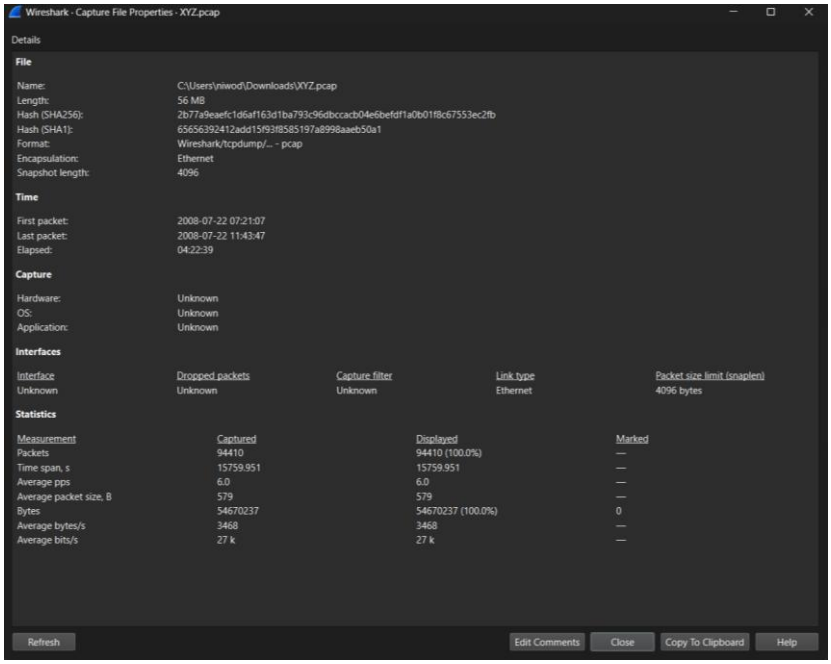

Figure 1. Capture File Properties of XYZ.pcap obtained from Wireshark 4.6.3

2.2 Network Components Identified

According to the capture summary as provided by Wireshark ver4.6.3 (Statistics → Endpoints), there were different Ethernet and IPv4 endpoints detected in the monitored dormitory network. The analysis was focused on identifying internal hosts operating within the 192.168.15.xx subnet, as this subnet corresponds to the dorm room referred to in the complaint (host 140.247.62.34\34.62.247.140.in-addr.arpa domain name pointer \G24.student.XYZ.org). The network components determined using the Ethernet Endpoints listed under Statistics are as below:

Device Role	IP Address	MAC Address	Manufacturer	Description
Suspect workstation	192.168.15.4	00:17:F2:E2:C0:CE	Apple Inc.	Apple laptop used by attacker
Gateway router	192.168.15.1	00:1D:D9:2E:4F:60	Hon Hai Precision (TRENDnet)	Dorm Wi-Fi router
External mail server	—	—	Google	mail.google.com

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Anonymous messaging service	—	—	willselfdestruct.com	Used to send harassing message

Table 1. Decomposition of different network component identified

Based on the statistics report, by analysing data from the ARP and Ethernet endpoints, the suspect host with IP address 192.168.15.4 matched the MAC address of 00:17:F2:E2:C0:CE, which is associated with Apple. Therefore, the suspect was using an Apple manufactured device, most likely a laptop. This further supported through analysis of endpoint statistics, which show that this device caused most of the HTTP traffic and that it generated the HTTP POST request used to deliver the malicious HTTP POST request.

The MAC address of the gateway device with the IP address 192.168.15.1 is also registered to Hon Hai Precision Industry (Foxconn) with ARP showing that the device was a TRENDnet router. This suggests that the dormitory network utilized a consumer TRENDnet Wi-Fi router, consistent with the scenario description. This router facilitated outbound traffic from the attacker device to external services including Gmail and willselfdestruct.com.

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3. Methodology

3.1 Tools Used

The Analysis of the packet capture file **XYZ.pcap** was conducted using the network forensic analysis tool named Wireshark. Wireshark was executed on a Windows-based forensic workstation and utilized for the inspection, filtering, and reconstruction of network communications across mutiple layers of the TCP/IP model.

The tool facilitated protocol-level analysis including Ethernet, IPv4, TCP, UDP, DNS, and SSL/TLS protocols. The analysis and filtering capabilities of the tool were utilized for the detection of active communication endpoints, filtering of malicious communications,isolate relevant network activity and reconstruct of application-layer exchanges using TCP stream analysis.

The external IP addresses observed within the capture file were resolved using standard DNS packet analysis within Wireshark. This enabled identification of external domains involved in the communications, including willselfdestruct.com and google services relevant to the investigation.

All analytical results were validated by reapplied filters and analysis of reconstructed TCP streams for consistency and reproducibility of results. No changes were made to the original packet capture file preseving intergrity.

3.2 Steps Involved

The packet capture archive was extracted and imported into Wireshark for detailed forensic examination. An initial assessment was conducted using the Statistics menu to assess endpoints and conversations at the Ethernet, IP, TCP, and UDP layers. This initial assessment showed several internal and external IP addresses engaged in network communications during the capture interval.

Endpoint statistics were subsequently evaluated to select internal hosts exhibiting a high level of network activity. This analysis revealed that IP address 192.168.15.4 to be the most active internal IP address, producing a significantly higher number of packets and bytes than other systems. This information led to a focus on communications related to this system.

Display filters were then used within Wireshark to selectively filter communications of interest and remove background noise. These filters, written in libpcap syntax, included protocol and address filters such as the following:

- tcp
- http
- dns
- ip.addr == 192.168.15.4
- ip.src == 192.168.15.4
- http.request.method == "POST"
- ip.addr == 192.168.15.1

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- frame contains "gausr="
- http.host contains "willselfdestruct"

The use of these filters enabled selective analysis of communications from the suspect system and through the network gateway.

Analysis of HTTP communications showed several web interactions, including form submissions. Special attention was paid to HTTP POST communications, which are indicative of user-submitted data.One POST communication was identified targeting /secure/submit on http://www.willselfdestruct.com

TCP stream reconstruction of this communication confirmed the presence of form data fields including the recipient email address,subject line and message body.The server response was observed as HTTP/1.1 302 Found with a redirect to/secure/success indicating successful processing of the submission.

3.3 Handling Data

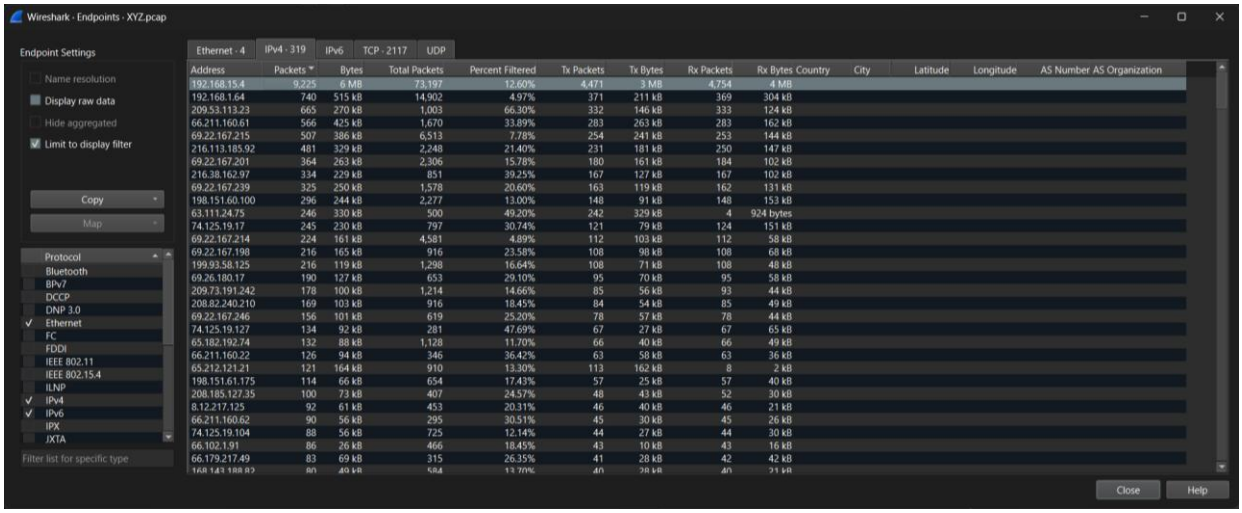

Figure 2. Protocol Hierarchy captured with Wireshark ver4.6.3

Packet Type	Number of packets	Requests	Technique	Responses	Technique
ARP	110	-	ARP	-	ARP
IP	73,197	-	-	-	-
TCP	70,713	Not calc.		Not calc.	
UDP	2,484	-	-	-	
DNS	2,322	-	DNS Query	-	DNS Response
HTTP	9,225	Present	http.request.method == "POST"	Present	HTTP Response

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SSL/TLS	1,153	-	TLS Handshake	-	TLS Handshake
Browser Data	697	-	Line-Based Text	-	-
Media/Images	4,929	-	PNG/JPG/GIF	-	HTTP Content
Data	57	-	Application Data	-	-

Table 2. Protocol Hirerarchy Packet distribution extracted from XYZ.pcap using Wireshark 4.63

The statistics in Table 2 show that the majority of the network traffic that was recorded was TCP-based, which is in line with typical web browsing behaviour. The internal IP address 192.168.15.4 was the most active host during the capture period, as evidenced by the significant volume of TCP request and response traffic displayed by Wireshark display filters. The traffic associated with 192.168.15.1 exhibited the expected behaviour of a gateway. HTTP communications, including numerous POST requests to external web services, contributed a significant portion of the activity. To ascertain their applicability to the inquiry, these POST requests underwent additional analysis. Section 5 of this report provides specifics on pertinent supporting evidence.

IP Address	As Sender	As Receiver
192.168.15.4	4,471	4,754
192.168.15.1	2,154	0
192.168.15.64	371	369
192.168.15.103	0	0
192.168.15.105	0	0

Table 3. Decomposition of IP traffic (Inbound & Outbound)

It is evident from the information presented in Table 2 that **192.168.15.4** was the most prominent contributor to the network activity captured during the analysis period, generating the highest volume of inbound and outbound traffic. Activity observed from **192.168.15.1** is consistent with its role as the network gateway device and can therefore be attributed to normal routing operations rather than user-driven behaviour.

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4. Detailed Findings

4.1 Important network players

No.	Entity	Role
01.	192.168.15.4	Suspect device responsible for Gmail login and message transmission
02.	192.168.15.1	Dormitory router forwarding outbound traffic
03.	jcoachj@gmail.com	Gmail account accessed by attacker
04.	lilytuckrige@yahoo.com	Victim email account
05.	willselfdestruct.com	Service used to transmit and destroy message
06.	Johnny Coach	Chemistry 109 student correlated with Gmail username

Table 4. Decomposition of Important network players

The investigation revealed several important network entities. The internal host 192.168.15.4 was identified as the attacker device, as it accessed the Gmail account jcoachj@gmail.com and subsequently transmitted the harassing message via willselfdestruct.com ARP and endpoint analysis confirmed that this IP address corresponds to a single Apple-manufactured device, indicating a unique workstation responsible for the malicious activity.

The IP address 192.168.15.1 was identified as the dormitory gateway router, forwarding outbound traffic from internal hosts to external services. MAC address resolution showed that this device is consumer-grade hardware manufactured by Hon Hai Precision Industry, consistent with the unsecured Wi-Fi router described in the scenario.

External services also played critical roles. The Gmail account jcoachj@gmail.com was actively used from the attacker device prior to message transmission, while willselfdestruct.com was utilized to submit and destroy the harassing message addressed to lilytuckrige@yahoo.com.

Correlation of the Gmail username with the Chemistry 109 class roster identified **Johnny Coach** as the only student whose name aligns with the account. Although network forensics cannot confirm physical keyboard use, the technical evidence demonstrates that the same device controlled the Gmail session and sent the malicious message, establishing strong attribution to Johnny Coach.

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4.2 Network Structure

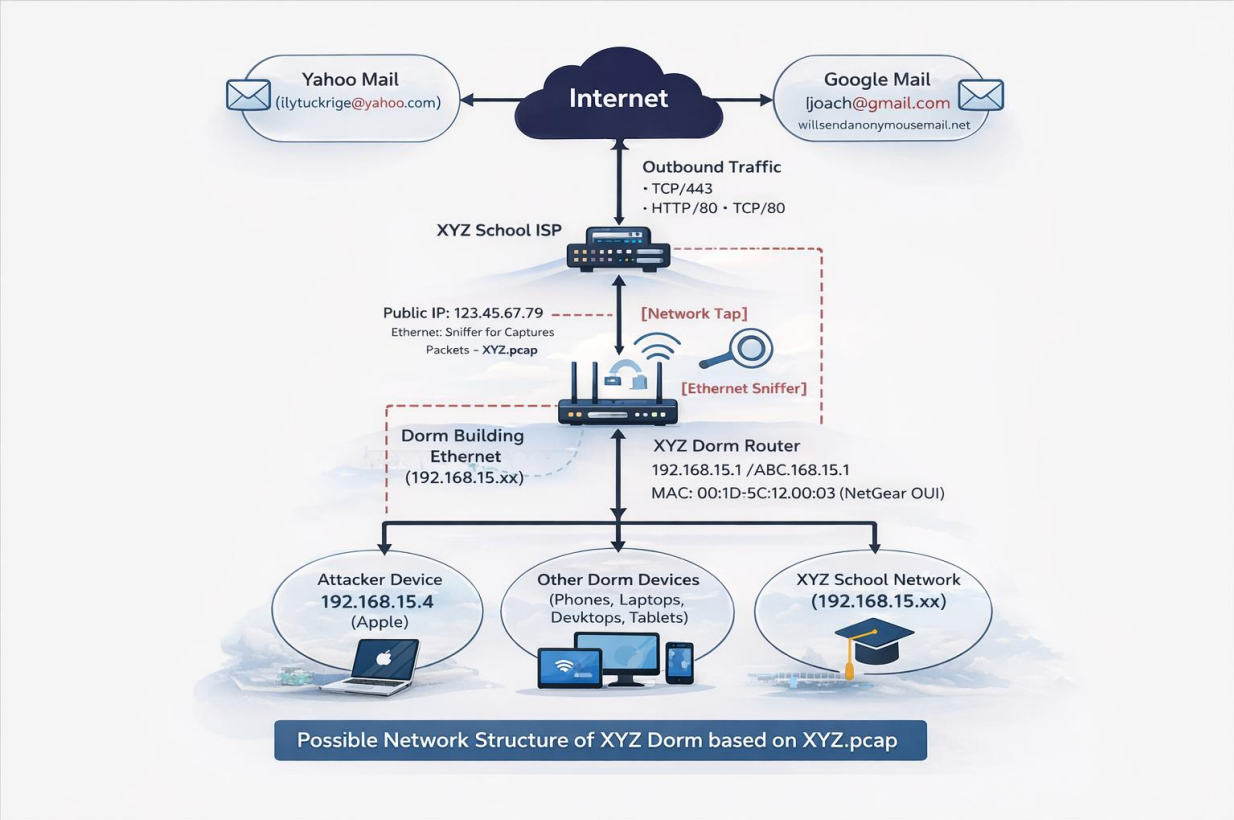

Figure 3. Possible Network Structure based on reconstruction from XYZ.pcap

The XYZ dormitory network consists of a shared Ethernet connection provided by the school, connected to a consumer-grade wireless router installed within the dorm room. This router, operating as the default gateway at IP address 192.168.15.1, provides Network Address Translation (NAT) and Internet access to multiple client devices through an unsecured Wi-Fi network. XYZ School IT deployed a network tap on the dormitory Ethernet port to capture all inbound and outbound traffic for forensic analysis.

Internally, devices operate within the private 192.168.15.0/24 subnet. The attacker device, identified as 192.168.15.4, communicated with the router for all external connections. Outbound traffic was forwarded through the router to XYZ School's ISP and onward to Internet-based services, including mail.google.com and willselfdestruct.com.

From a logical perspective, the router acts as the central aggregation point, relaying traffic between internal hosts and external services while masking individual devices behind a shared public IP address. This topology explains why initial email headers only revealed a dormitory-level IP, requiring packet-level analysis to identify the specific internal host responsible for the malicious activity.

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4.3 Activity Timeline for 192.168.15.4

Packet Number	Activity	Destination	Inference
77520	TCP session established with google	74.125.19.104	Suspect initiated by external web browsing
77528	GET request to google resource	74.125.19.104	Active google session detected
77520-77535	Cookie exchange (OL_SESSION=jcoachj@gmail.com)	74.125.19.104	User authenticated as jcoachj@gmail.com
80611	Destination Domain : willselfdestruct.com	69.80.225.91	Connection established to anonymous email site
80613	POST path: /secure/submit	69.80.225.91	Continued interaction with external web service
80613	Form data transmitted (recipient, subject, message)	69.80.225.91	Connection established to anonymous email site
80615	HTTP/1.1 302 Found – Redirect to /secure/success	69.80.225.91	Server confirms successful submission
82423–82428	Additional HTTP responses from server	69.80.225.91	Session closed after submission

Table 5. Activity Timeline based on network forensic analysis and event reconstruction

Note: Only distinct IP addresses with different time stamps are mentioned on the table. Each IP has a sequence of following TCP/HTTP packets following it which is not captured in this table. Packet numbers provided in the table indicate the first occurrence of the transaction.

4.4 Background Evidence

1. The network capture file XYZ.pcap was checked for forensic integrity by using the cryptographic hash values. The MD5 and SHA1 hash values for the uncompressed capture file were calculated and checked to ensure that they matched the values provided in Section 2.1 of this report. This ensures that the evidence that was analysed was not altered during the entire process of the investigation.
2. All the communications that were captured took place within the time frame described in Section 2.1 (Network Capture File Details). This analysis only provides information on the communications that took place during this time frame.
3. Analysis of the endpoints by Statistics → Endpoints → IPv4 in Wireshark revealed the presence of several active devices on the internal 192.168.15.xx network. Among these, the internal gateway device was found to be 192.168.15.1, which acted as the network router. The most active internal device was 192.168.15.4, which transmitted and received the largest number of packets during the capture period.

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4. ARP traffic analysis revealed that other internal devices had little activity and did not contribute meaningfully to the communications related to the incident. Filtering by:
   - **ip.src == 192.168.15.X**
   - **ip.dst == 192.168.15.X**

   Revealed no significant packet exchanges from other hosts. Therefore, these devices were considered unrelated to the incident.

5. DNS analysis using the filter
   - **dns && ip.addr == 192.168.15.4**

   Revealed multiple requests for resolution of domains coming from the suspect device. The DNS traffic revealed requests for external web services such as Google and willselfdestruct.com used to send the harassing message. This is an indication of active web browsing by the user of 192.168.15.4

6. Further HTTP traffic analysis using:

   - **http && ip.addr == 192.168.15.4**

   Revealed several HTTP GET and POST requests. Of particular note was the HTTP POST request to http://www.willselfdestruct.com with the path /secure/submit which included form data such as:

   - Recipient email address: lilytuckrige@yahoo.com
   - Subject line
   - Message body containing harassing language

7. **The Follow TCP Stream** function in Wireshark was used to recreate the entire unencrypted HTTP transmission. This verifies that the email was sent over an unencrypted HTTP connection. The server response indicated an HTTP/1.1 302 redirect to /secure/success, indicating that the submission was successfully processed by the web server.

8. Analysis of HTTP headers and cookies revealed an authenticated session belonging to the Google account jcoachj@gmail.com. The existence of this session during the time of the email transmission provides a strong connection between the suspect device and the discovered user account.

9. User-agent analysis of the HTTP requests revealed that the device was running a desktop computer system with a standard web browser. The web browsing activity observed is consistent with user-agent activity and not automated background processes.

10. No evidence was found to suggest the existence of phishing infrastructure, tunnelling, firewall modification, malicious software downloads, or coordinated accomplice communication. All observed activity is consistent with independent web browsing activity

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leading to the eventual submission of a harassing message via an anonymous web-based email service.

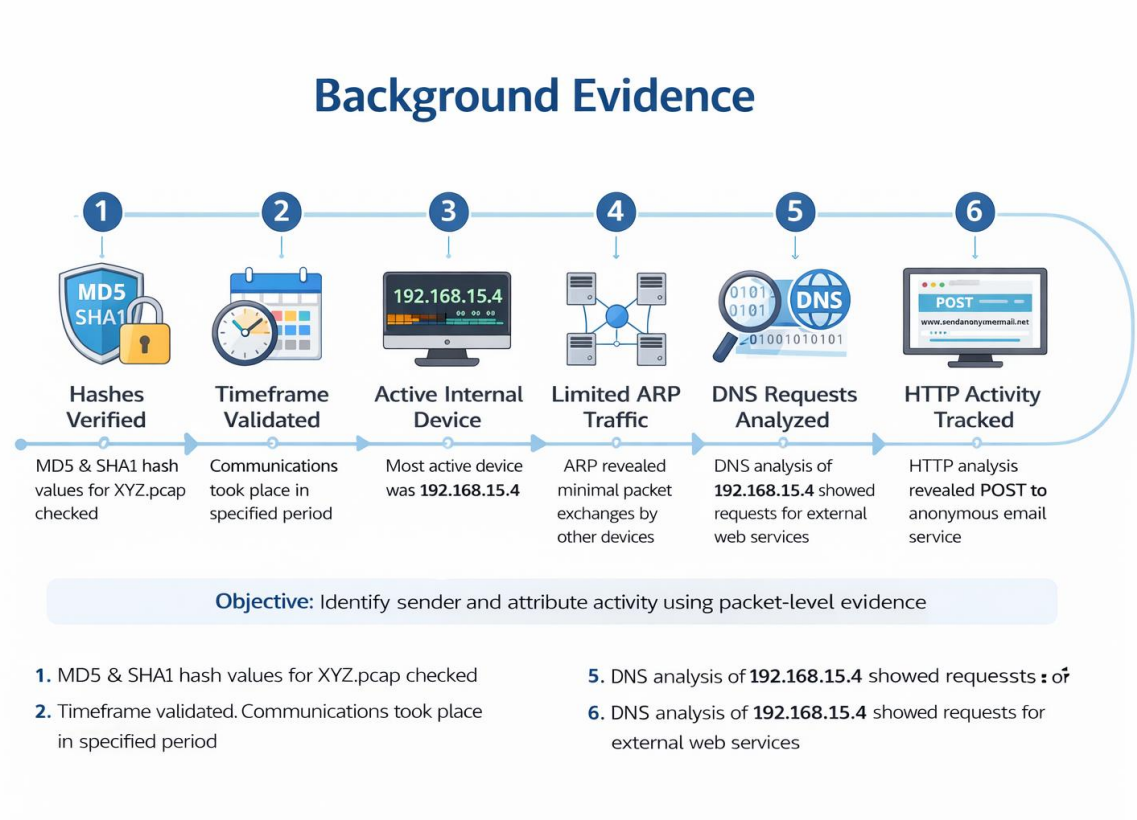

Figure 4. Background Evidence

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5. Supporting Evidence Presented

Evidence ID	Evidence Description	Source / Tool	Key Finding	Relevance to Case
E1	Hash verification of XYZ.pcap (MD5, SHA1, SHA256)	Provided scenario / Wireshark	File integrity confirmed	Ensures evidence was not altered
E2	Capture File Properties summary	Wireshark – Statistics → Summary	94,410 packets captured between 22-JUL-2008 07:21:07 and 11:43:47	Establishes investigation timeframe
E3	ARP reply: "192.168.15.1 is at 00:1D:D9:2E:4F:60"	Wireshark – ARP filter	Router MAC identified (Hon Hai Precision)	Confirms gateway device
E4	ARP mapping for 192.168.15.4 → 00:17:F2:E2:C0:CE	Wireshark – ARP filter	Attacker MAC identified (Apple Inc.)	Links IP to physical device
E5	IPv4 Endpoint statistics	Wireshark – Statistics → Endpoints	192.168.15.4 generated majority of HTTP traffic	Identifies primary suspect host
E6	HTTP session to mail.google.com	Wireshark – Follow TCP Stream	Gmail login from 192.168.15.4 (jcoachj@gmail.com)	Shows account control
E7	HTTP POST /secure/submit to willselfdestruct.com	Wireshark – TCP Stream Reconstruction	Harassing message payload recovered	Direct proof of malicious action
E8	HTTP 302 redirect to /secure/success	Wireshark	Confirms successful submission of message	Proves message transmission completed
E9	Timeline correlation (Gmail login → POST)	Packet timestamps	Sequence of events reconstructed	Demonstrates intentional behaviour
E10	Email header showing public IP 140.247.62.34	Provided screenshot	Attack originated from dorm network	Establishes geographic origin
E11	Chemistry 109 class roster	Provided scenario	"Johnny Coach" matches jcoachj@gmail.com	Human attribution correlation
E12	OUI lookup for MAC 00:17:F2	Wireshark vendor resolution	Apple device identified	Confirms attacker device type
E13	OUI lookup for MAC 00:1D:D9	Wireshark vendor resolution	Hon Hai Precision (router hardware)	Confirms router manufacturer

Table 6. Tabulated list of evidence supporting the Forensic report

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6. Conclusions

This forensic analysis examined the network capture file XYZ.pcap to see if the device with the IP address 192.168.15.4 was involved in the transmission of a harassing email through an anonymous email service. The forensic analysis was performed using Wireshark protocol filters, HTTP analysis, TCP stream analysis, and the correlation of session artifacts.

The IPv4 endpoint statistics confirmed that 192.168.15.4 was one of the busiest internal devices during the time of the capture. The device was involved in heavy TCP traffic, which was mostly HTTP traffic to external web services. There was no unusual routing activity or IP spoofing detected. With the ip.addr == 192.168.15.4 filter, heavy web traffic was detected. Further filtering with http.request.method == "POST" showed that several POST requests had been made from this device. Among them, a significant POST request was detected. The request targeted the resource /secure/submit, which meant that the request was transmitting form data.

Analysis of the pertinent TCP flow (Stream 1637) showed the complete HTTP request, which included:

- Recipient's email address: lilytuckrige@yahoo.com
- Subject: Your class stinks
- Body text with harassing content
- Typical browser user agent string
- Valid HTTP headers and session cookies

The server response observed was HTTP/1.1 302 Found with a redirect to /secure/success .This response indicates that the request was successfully handled by the remote server indicating that the email was successfully sent and not just attempted.

Further analysis of the HTTP cookies showed the presence of an active session with the ID OL_SESSION=jcoachj@gmail.com. This demonstrates that, at the time of sending the anonymous email, the device was operating under the control of an authenticated Google session for the Gmail account jcoachj@gmail.com. There are no indicators in the capture file that the system was compromised, remotely controlled, or used as a proxy for malicious actors. The flow indicates typical manual browsing activity, including page requests, form submissions, data submission, and server responses.

The chronological activity order from the timeline is as follows:

1. Active Google browsing session
2. Gmail authentication session maintained
3. Navigation to willselfdestruct.com
4. HTTP POST submission to /secure/submit
5. Successful server response HTTP 302 redirect confirming processing

This chronological order of activities adds to the attribution, as it demonstrates that the activities were structured and user-driven consistent with manual web browsing and form submission rather than automated or background system processes.

On the basis of the cumulative network evidence, including IP address correlation, HTTP POST analysis, TCP stream analysis, form data recovery, and authenticated session artifacts, this

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report concludes that the device with IP address 192.168.15.4 was responsible for sending the anonymous harassing email during the recorded period. It is recommended that the organizational records be searched to determine which individual was assigned to this device during the time of the incident. Further proceedings can be initiated in accordance with the organizational conduct policy and cybercrime laws that regulate online harassment and the misuse of communication platforms.

The results in this report are verified by packet-level forensic data and can be reproduced using the filters and stream analysis techniques mentioned in the report.

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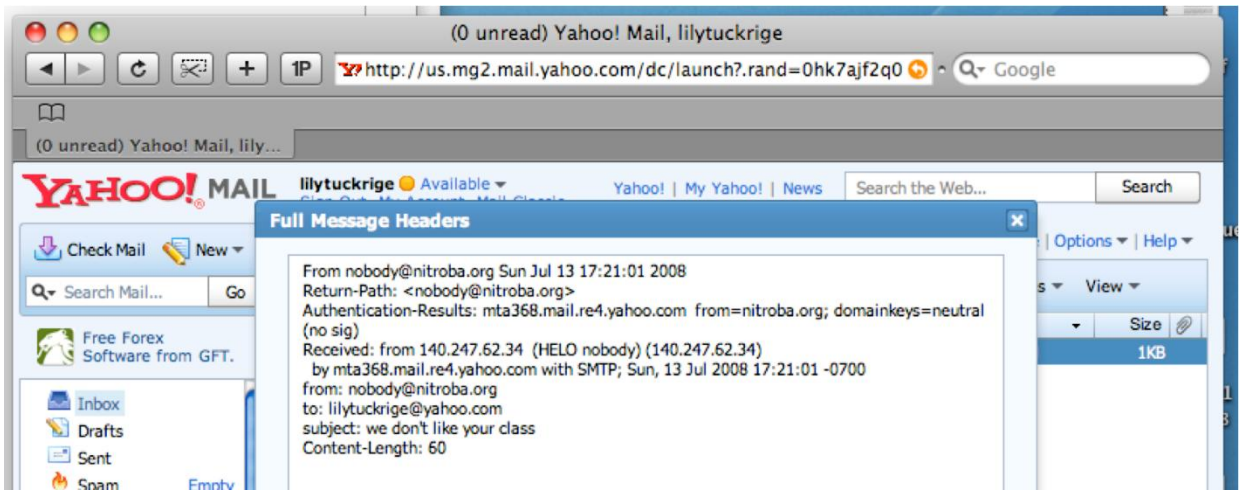
7. Appendix A – List of figures

Evidence 014



Description: This figure illustrates the full email message headers obtained from Lily Tuckrige's Yahoo Mail account. The headers reveal that the harassing email was received from the internal IP address 140.247.62.34, which corresponds to the XYZ School dormitory network. The header also records the sender domain (nitroba.org) and delivery path through Yahoo mail servers, providing initial attribution at the network level. This public IP establishes the dormitory network as the source location, justifying the deployment of a packet capture device for deeper internal forensic analysis.

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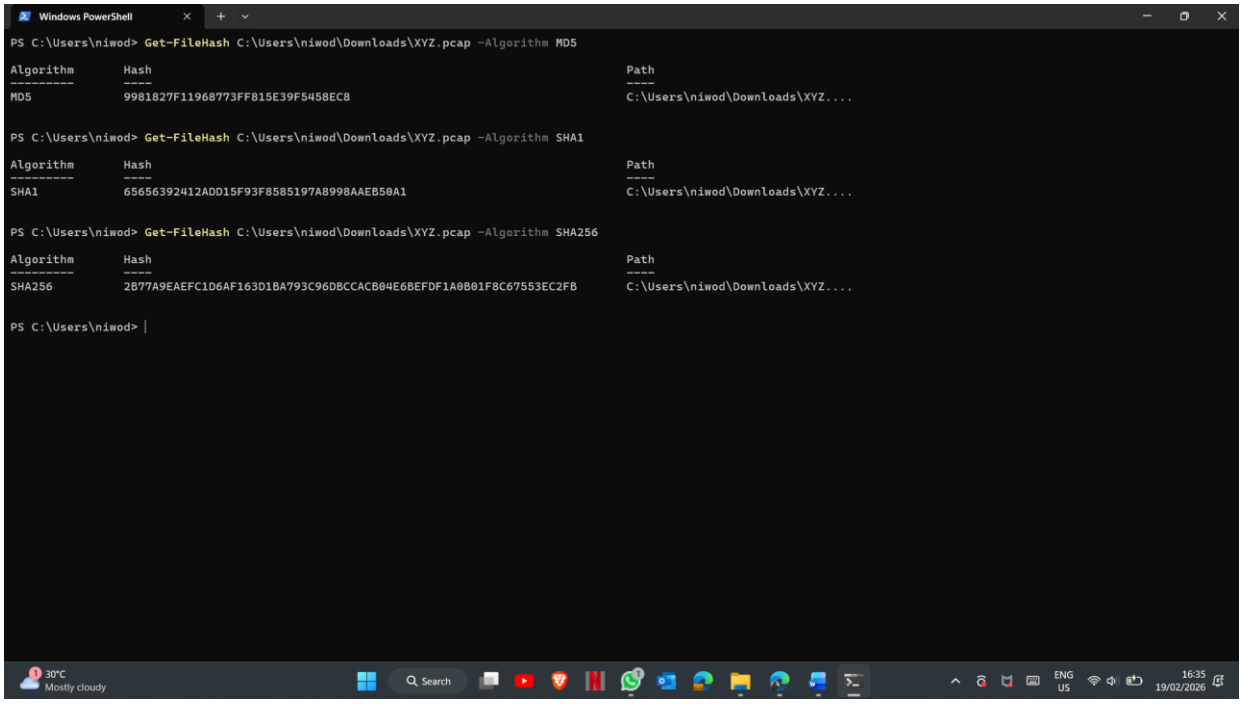
Evidence 015



```
Windows PowerShell

PS C:\Users\niwod> Get-FileHash C:\Users\niwod\Downloads\XYZ.pcap -Algorithm MD5

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             9981827F11968773FF815E39F5458EC8                                       C:\Users\niwod\Downloads\XYZ....

PS C:\Users\niwod> Get-FileHash C:\Users\niwod\Downloads\XYZ.pcap -Algorithm SHA1

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA1            65656392412ADD15F93F8585197A8998AAEB50A1                               C:\Users\niwod\Downloads\XYZ....

PS C:\Users\niwod> Get-FileHash C:\Users\niwod\Downloads\XYZ.pcap -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          2B77A9EAEFC1D6AF163D1BA793C96DBCCACB04E6BEFDF1A0B01F8C67553EC2FB       C:\Users\niwod\Downloads\XYZ....

PS C:\Users\niwod>
```

Description: This figure shows the MD5, SHA1, and SHA256 hash values generated for the XYZ.pcap file using Windows PowerShell. These cryptographic hashes confirm the integrity of the packet capture and ensure that the evidence was not altered during analysis. Matching hash values validate the authenticity of the forensic dataset used throughout this investigation.

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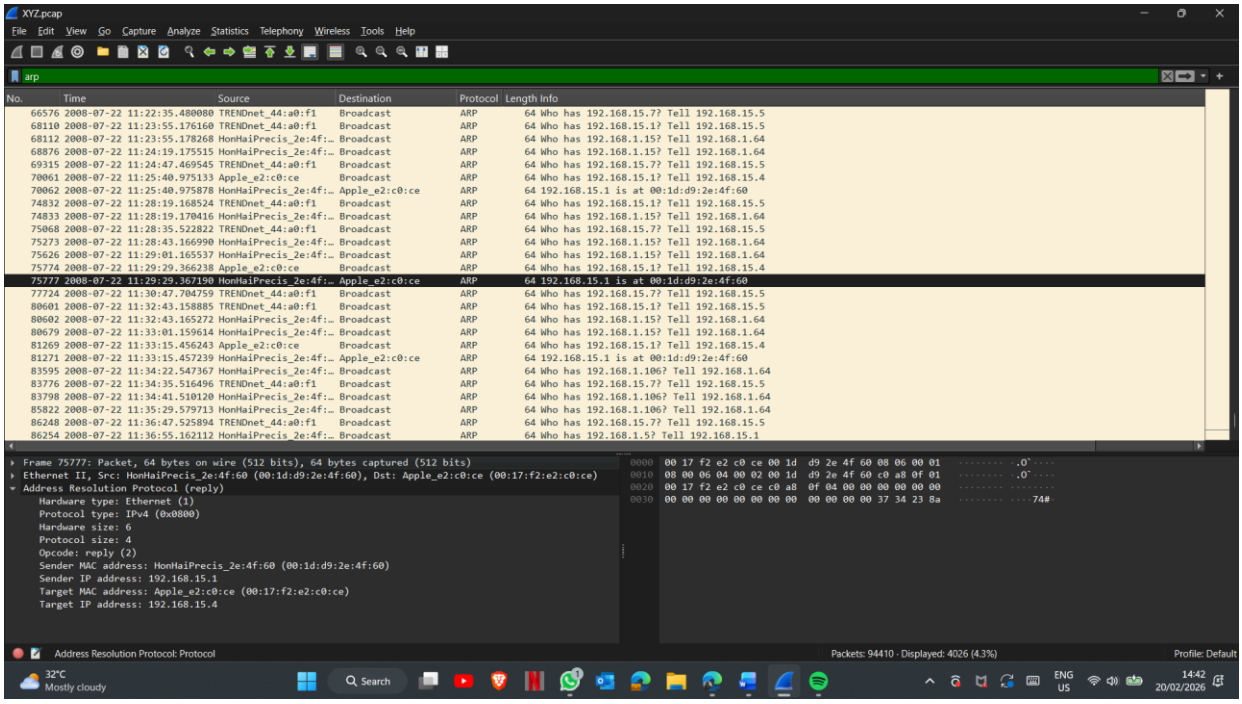
Evidence 016



Description: This figure shows ARP traffic from the captured network, specifically the reply stating "192.168.15.1 is at 00:1D:D9:2E:4F:60." This confirms the MAC address of the dormitory gateway router. The ARP exchange also identifies the attacker device (192.168.15.4) through its Apple-associated MAC address, establishing IP-to-physical device mapping critical for attribution.

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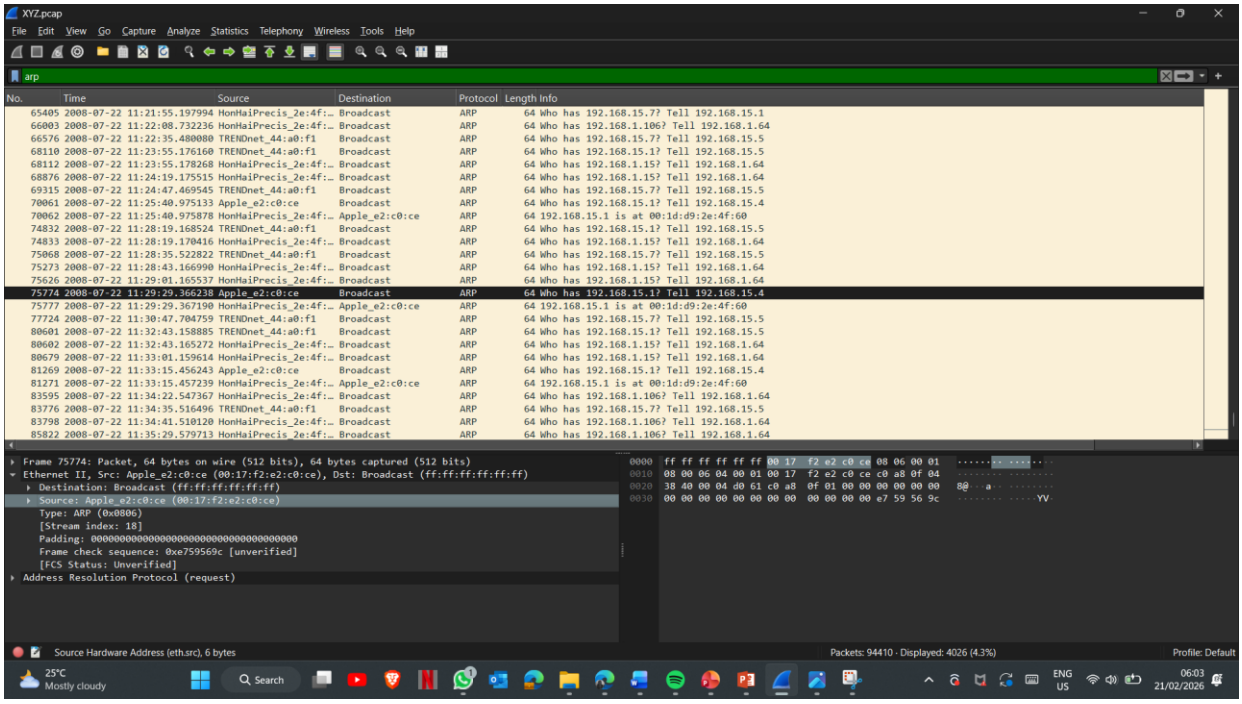
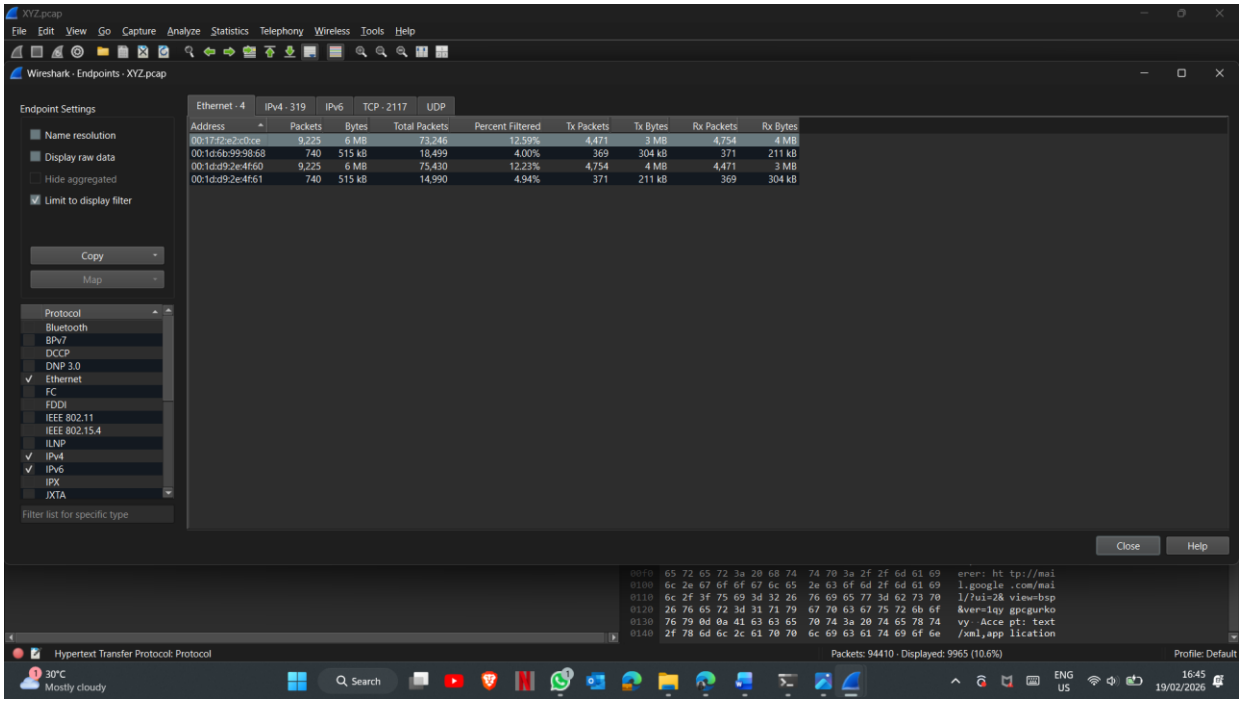
Evidence 017





Description: These figures show an ARP request originating from the Apple device (MAC 00:17:F2:E2:C0:CE), corresponding to IP address 192.168.15.4. The request asks for the hardware address of 192.168.15.1, confirming active communication between the attacker's device and the dormitory router. This further validates the attacker's dependency on the router for external network access.

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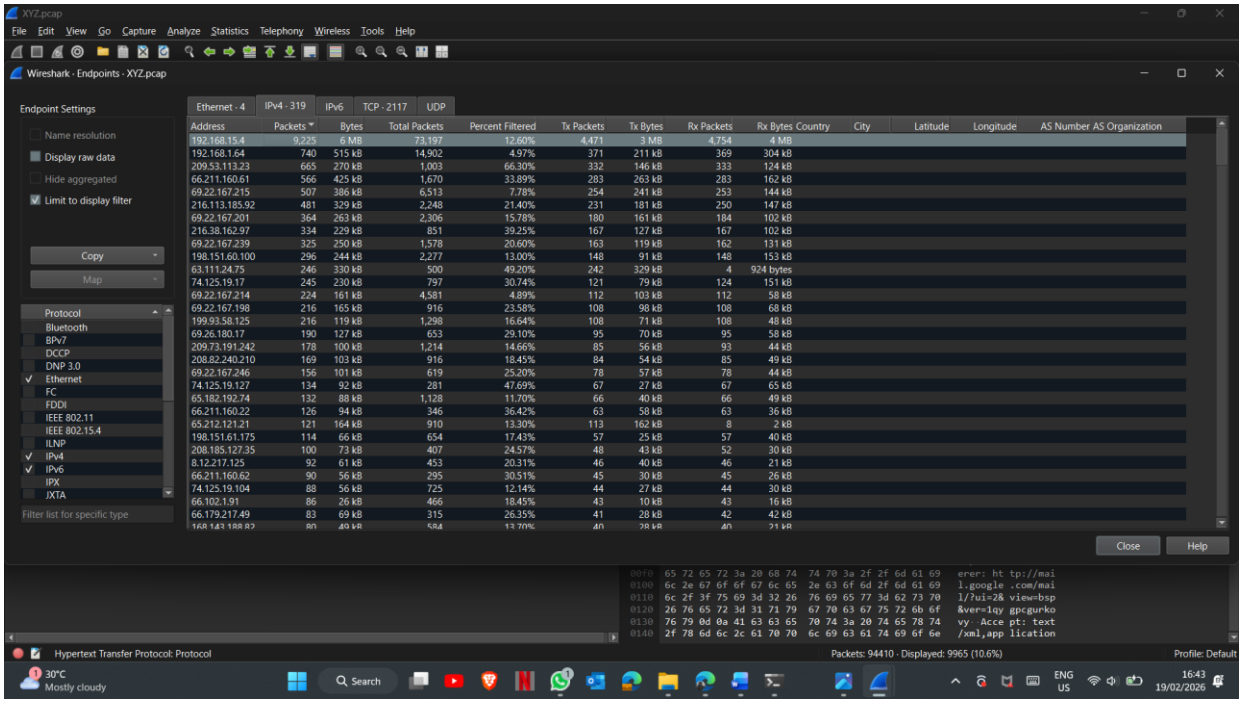
Evidence 018



Description: This figure displays the IPv4 endpoint statistics extracted from XYZ.pcap. The internal host 192.168.15.4 generated a significant volume of traffic compared to other private addresses, indicating active user behaviour. The router at 192.168.15.1 is also visible as the central gateway. This endpoint analysis supports identification of 192.168.15.4 as the primary suspect device.

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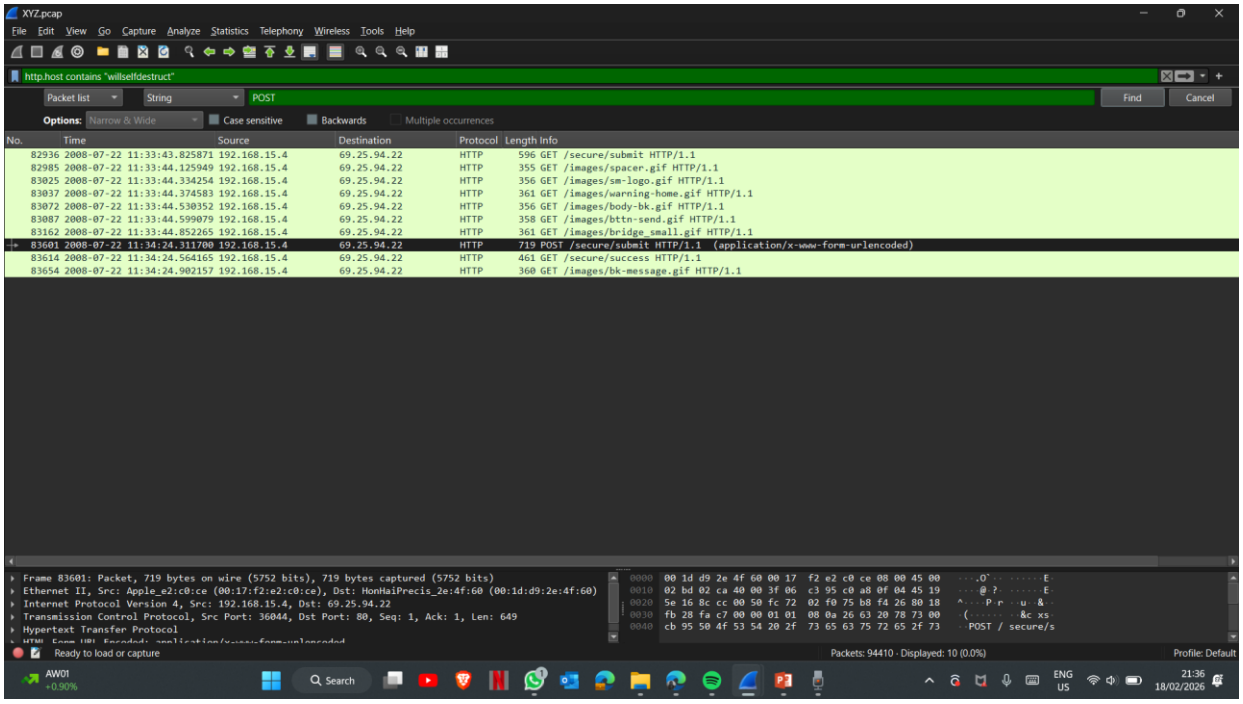
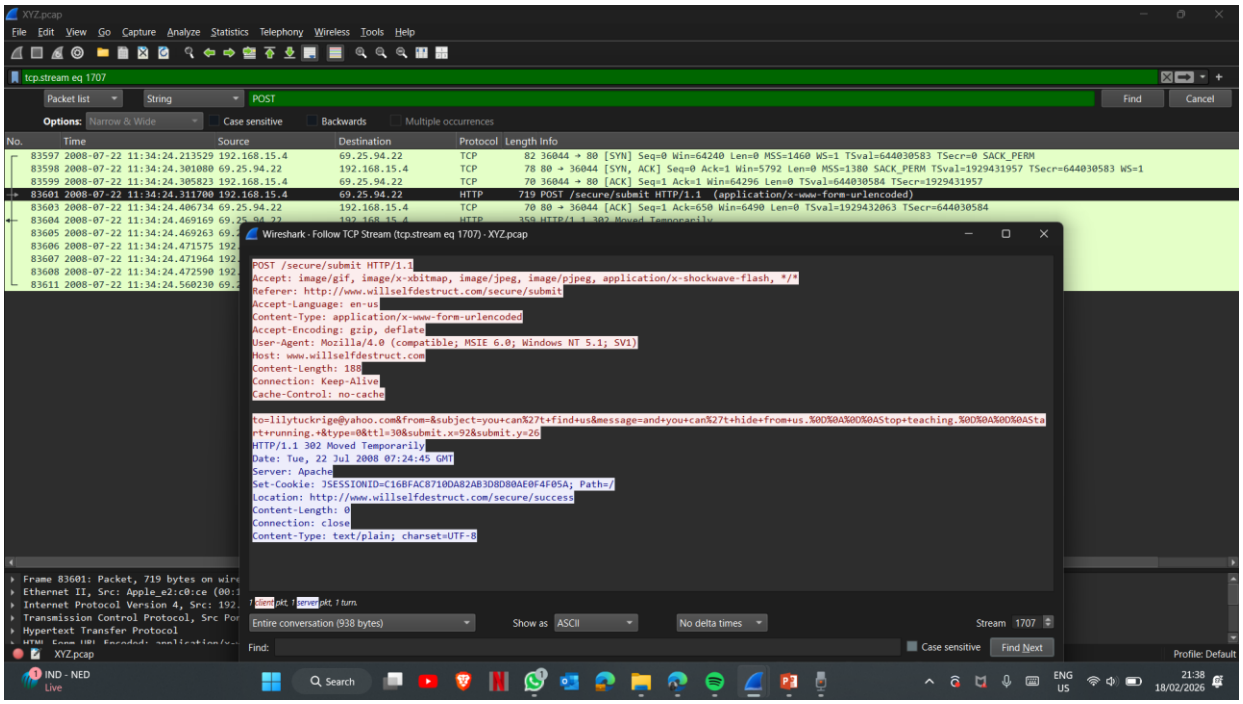
Evidence 019





Description: These figures show the reconstructed HTTP POST request sent from the attacker device (192.168.15.4) to willselfdestruct.com, containing the destination address lilytuckrige@yahoo.com and the harassing message content. The associated TCP stream further confirms successful delivery through an HTTP 302 redirect to "/secure/success." This evidence conclusively demonstrates that the malicious message originated from the suspect device and was transmitted intentionally via the willselfdestruct service.

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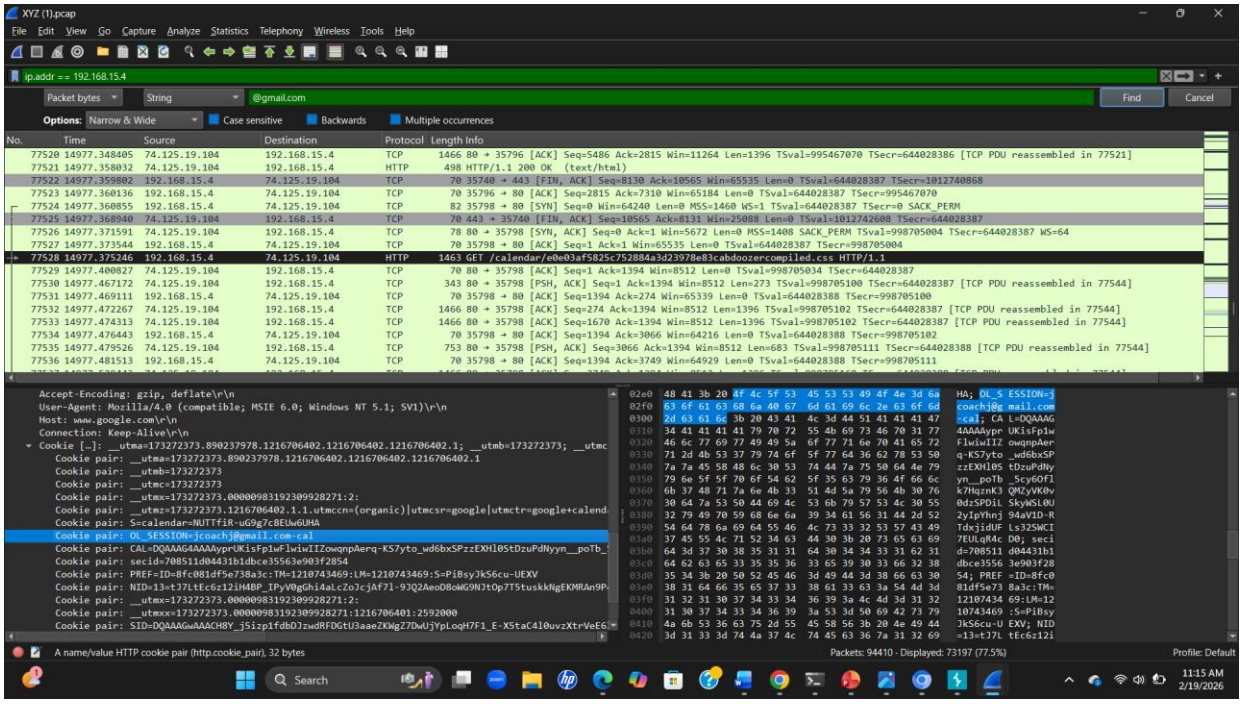
Evidence 020



Description: This figure shows HTTP traffic between the suspect device (192.168.15.4) and mail.google.com, including multiple Gmail-related cookies and session identifiers. The captured packets confirm active Gmail usage originating from the attacker's IP address, supporting reconstruction of the Gmail session associated with jcoachj@gmail.com. This evidence demonstrates that the same device responsible for the willselfdestruct submission was also actively accessing Gmail during the investigation period.

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8. Self-review section

WD Ashmitha (Student ID: s8200500 / 30971)

This investigation was done collaboratively, with well-defined responsibilities. My main efforts were in the preparation of the Executive Summary, Methodology, Background Evidence, Activity Timeline, and Conclusions. These sections demanded in-depth packet-level analysis and correlation of technical artefacts.

By doing a structured analysis of the capture file, we were able to find that the device 192.168.15.4 sent an HTTP POST request to willselfdestruct.com with the harassing message, and an authenticated Gmail session was also active during the same period. The results obtained by IP analysis, MAC mapping, session artefacts, and timeline analysis helped to reinforce device-level attribution.

This investigation has again emphasised the need for evidence integrity, filtering, and careful interpretation of results. Although network forensics cannot prove physical user interaction, the results obtained are based on reproducible technical analysis.

SKK Niwodhya Gunasena (Student ID: s8200495 / 30941)

As part of the thorough examination of packet captures and email headers, I performed a network forensic investigation on the XYZ network to determine who sent the unsolicited harassing email messages to Lily Tuckrige. Using the Wireshark program, I was able to determine that the network traffic captured in the file XYZ.pcap could be filtered based on IP address information and then analysed to isolate suspicious activity from internal users. By filtering out all other internal users, I identified an "unknown user" on the IP address 192.168.15.4, as well as a dormitory router at the IP address of 192.168.15.1, as originating from a network in the dormitory and sending e-mail to lilytuckrige@yahoo.com.

By performing an ARP analysis of these devices, I was able to determine that the "unknown user" was using a Macintosh computer. In addition, by reconstructing the TCP streams of the suspect's communications, I was able to determine the suspect was accessing Gmail through jcoachj@gmail.com and identified the suspect's activity relating to submitting anonymous e-mail messages through HTTP POST requests (to willselfdestruct.com) as being sent to lilytuckrige@yahoo.com. The time correlation between accessing Gmail and sending an anonymous communication to lilytuckrige@yahoo.com (using the "unknown user") supports that the same "unknown user" accessed both services from a single device. Even with many of these limitations, I reconstructed the anonymous email; identified the device initiating the email; and believed with confidence that Johnny Coach was the person responsible for the email when I associated the Gmail account to the Chemistry 109 student list. Future investigations would benefit from improving internal logging, implementing better network segmentation, and capturing data for longer periods of time, all of which would improve the accuracy of attribution.

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